Fig 1. IT1 · SQL injection: search x' OR '1'='1 returns no patients

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Register new patient

First name *

Last name *

Date of birth * 2000 / 01 / 01

Gender * M

Phone * (10 digits)

Email

Blood group Choose an option

Address

Register patient

Book appointment

Search patient (name or phone)

x' OR '1'='1

No matching patients. Register one first.

Upcoming appointments

appt_id	appt_date	appt_time	patient	doctor	status
16	2026-10-19	10:00	Ravi Kumar	Dr. Rahul Verma	Scheduled
11	2026-10-20	10:00	Nisha Joseph	Dr. Arvind Menon	Scheduled
12	2026-10-20	11:00	Vijay Anand	Dr. Rahul Verma	Scheduled

Fig 2. IT2 · Full journey (1/6): new patient registered, Patient ID = 36

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Register new patient

First name *

Last name *

Date of birth *

2000 / 01 / 01

Gender *

M

Phone * (10 digits)

Email

Blood group

Choose an option

Address

Register patient

Patient registered. Patient ID = 36

Book appointment

Search patient (name or phone)

Patient

12 - Vijay Anand (9840020012)

Department

Cardiology

Doctor

Dr. Meera Iyer (fee 850.00)

Date

2026 / 10 / 19

Show booked slots too (to demonstrate the double-booking error)

Time slot

09:00

Reason for visit

Book appointment

Fig 3. IT2 · Full journey (2/6): appointment booked with sp_book_appointment, ID = 171

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
1	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Register new patient

First name *

Last name *

Date of birth * 2000 / 01 / 01

Gender * M

Phone * (10 digits)

Email

Blood group Choose an option

Address

Register patient

Book appointment

Search patient (name or phone) 9876500001

Patient 36 - Test Journey (9876500001)

Department Cardiology

Doctor Dr. Meera Iyer (fee 850.00)

Date 2026 / 10 / 19

Show booked slots too (to demonstrate the double-booking error)

Time slot 09:00

Reason for visit Integration test

Book appointment

Appointment booked. Appointment ID = 171

Fig 4. IT2 · Full journey (3/6): Doctor Desk - prescription added, 2 lab tests ordered

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

5 − +

Notes

Add to prescription

medicine	dosage	frequency	days	notes
Amlodipine	1 tablet	once daily	5	None

Order lab tests

Tests

Choose options

Order selected tests

Ordered: Total Cholesterol

Ordered: Hemoglobin

test	status	result	unit	cost
Hemoglobin	Ordered	None	g/dL	150
Total Cholesterol	Ordered	None	mg/dL	300

Fig 5. IT2 · Full journey (4/6): Generate invoice refused - ERR_LABS_PENDING

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	6	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Billing Desk | Analytics

Visit to bill

#171 Test Journey with Dr. Meera Iyer (19 Oct 09:00) [Not billed, 2 lab pending]

Only visits ready to bill

Test Journey

Cardiology · Dr. Meera Iyer · visit #171 on 19 Oct 2026 at 09:00

Item	Status	Amount (₹)
Consultation - Dr. Meera Iyer	-	850.00
Lab - Total Cholesterol	Ordered	300.00
Lab - Hemoglobin	Ordered	150.00

Expected total: ₹1,300.00

2 lab test(s) are not completed yet. The procedure will refuse to invoice this visit until the lab finishes. You can press Generate to show this.

Invoice

Generate invoice

Complete all lab tests before generating the invoice

No invoice yet. Generating one sums the consultation fee and all lab tests in a single transaction.

Recently paid

bill	Paid at	patient	doctor	Consultation ₹	Labs ₹	Total ₹	payment_mode
1	18 Oct, 18:50	Ravi Kumar	Dr. Arvind Menon	400.00	300.00	700.00	UPI
2	18 Oct, 17:50	Divya Sharma	Dr. Arvind Menon	400.00	0.00	400.00	Card

Fig 6. IT2 · Full journey (5/6): labs completed, invoice generated for Rs 1,300

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	2

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Billing Desk | Analytics

Invoice #157 generated for Test Journey: ₹1,300.00

Visit to bill

#171 Test Journey with Dr. Meera Iyer (19 Oct 09:00) [Not billed]

Only visits ready to bill

Test Journey

Cardiology · Dr. Meera Iyer · visit #171 on 19 Oct 2026 at 09:00

Item	Status	Amount (₹)
Consultation - Dr. Meera Iyer	-	850.00
Lab - Total Cholesterol	Completed	300.00
Lab - Hemoglobin	Completed	150.00

Expected total: ₹1,300.00

Invoice

Regenerate invoice

Amount due

₹1,300.00

Payment mode

Cash

Mark as paid

Download invoice (.txt, unpaid)

Recently paid

bill	Paid at	patient	doctor	Consultation ₹	Labs ₹	Total ₹	payment_mode

Fig 7. IT2 · Full journey (6/6): paid by UPI - bill Paid, visit Completed

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Billing Desk | Analytics

Payment of ₹1,650.00 received by UPI. Bill #157 is Paid and visit #171 is Completed. Download the paid invoice under Recently paid.

Visit to bill

#171 Test Journey with Dr. Meera Iyer (19 Oct 09:00) [Generated]

Only visits ready to bill

Ravi Kumar

Cardiology · Dr. Rahul Verma · visit #16 on 19 Oct 2026 at 10:00

Item	Status	Amount (₹)
Consultation - Dr. Rahul Verma	-	900.00
Lab - Fasting Blood Glucose	Ordered	80.00
Lab - Total Cholesterol	Sample Collected	300.00
Lab - HDL Cholesterol	Ordered	350.00

Expected total: ₹1,630.00

3 lab test(s) are not completed yet. The procedure will refuse to invoice this visit until the lab finishes. You can press Generate to show this.

Invoice

Generate invoice

No invoice yet. Generating one sums the consultation fee and all lab tests in a single transaction.

Recently paid

Fig 8. IT3 · Test ordered after invoicing: bill back to Draft, provisional total Rs 1,650 (trigger)

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	5	2

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Billing Desk | Analytics

Visit to bill

#171 Test Journey with Dr. Meera Iyer (19 Oct 09:00) [Draft, 1 lab pending]

Only visits ready to bill

Test Journey

Cardiology · Dr. Meera Iyer · visit #171 on 19 Oct 2026 at 09:00

Item	Status	Amount (₹)
Consultation - Dr. Meera Iyer	-	850.00
Lab - Total Cholesterol	Completed	300.00
Lab - Hemoglobin	Completed	150.00
Lab - HDL Cholesterol	Ordered	350.00

Expected total: ₹1,650.00

1 lab test(s) are not completed yet. The procedure will refuse to invoice this visit until the lab finishes. You can press Generate to show this.

Invoice

A test was ordered after the invoice was made, so the bill went back to Draft. Regenerate it to pick up the new total.

Regenerate invoice

Amount due (provisional)

₹1,650.00

Payment opens once the invoice is regenerated.

Recently paid

bill	Paid at	patient	doctor	Consultation ₹	Labs ₹	Total ₹	payment_mode
1	18 Oct, 18:50	Ravi Kumar	Dr. Arvind Menon	400.00	300.00	700.00	UPI

Fig 9. IT4 · Ordering a test on a paid visit is refused (trg_block_lab_order_after_payment)

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Dosage (e.g. 1 tablet)

Frequency (e.g. twice daily)

Duration (days) 5 − +

Notes

Add to prescription

medicine	dosage	frequency	days	notes
Amlodipine	1 tablet	once daily	5	None

Order lab tests

Tests

Choose options

Order selected tests

ESR: Cannot add lab tests to an appointment whose bill is paid

test	status	result	unit	cost
Hemoglobin	Completed	14	g/dL	150
Total Cholesterol	Completed	250	mg/dL	300
HDL Cholesterol	Completed	50	mg/dL	350

Fig 10. IT5 · Appointment cancelled at Reception

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Time slot

09:30

Reason for visit

Integration test

Book appointment

Upcoming appointments

appt_id	appt_date	appt_time	patient	doctor	status
171	2026-10-19	09:00	Test Journey	Dr. Meera Iyer	Completed
16	2026-10-19	10:00	Ravi Kumar	Dr. Rahul Verma	Scheduled
11	2026-10-20	10:00	Nisha Joseph	Dr. Arvind Menon	Scheduled
12	2026-10-20	11:00	Vijay Anand	Dr. Rahul Verma	Scheduled
13	2026-10-21	09:00	Pooja Krishnan	Dr. Anita Desai	Scheduled
14	2026-10-21	16:00	Imran Khan	Dr. Priya Nair	Scheduled
15	2026-10-22	12:00	Deepa Venkat	Dr. Lakshmi Pillai	Scheduled

Cancel an appointment

#16 - Ravi Kumar with Dr. Rahul Verma on 2026-10-19 10:00

Cancel selected appointment

Appointment cancelled (the audit trigger logs this in Audit_Logs).

Fig 11. IT5 · Audit trail shows the CANCELLED row (trg_audit_appointment_cancellation)

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).





Peak hour: **10:00** with 15 appointments.

› Show SQL

› Show SQL

Doctor workload and revenue

doctor	department	visits	Revenue ₹	₹ per visit
Dr. Meera Iyer	Cardiology	7	10100	1443
Dr. Rahul Verma	Cardiology	7	7230	1033
Dr. Vikram Singh	Orthopedics	6	7070	1178
Dr. Arvind Menon	General Medicine	9	5870	652
Dr. Lakshmi Pillai	Dermatology	6	5720	953
Dr. Sunita Rao	General Medicine	5	5230	1046
Dr. Priya Nair	Orthopedics	5	4180	836
Dr. Farhan Sheikh	Dermatology	5	4000	800
Dr. Anita Desai	Pediatrics	4	3750	938
Dr. Karthik Subramanian	Pediatrics	5	3700	740

› Show SQL

Payment modes



› Show SQL

⌄ Audit trail (written by trg_audit_appointment_cancellation)

log_id	logged_at	appt_id	action	old_status	new_status	details
16	2026-10-19 19:43:25	16	CANCELLED	Scheduled	Cancelled	Doctor 3, patient 1, slot 2026-10-19 10:00:00
15	2026-10-19 19:42:56	171	STATUS_CHANGE	Scheduled	Completed	Doctor 4, patient 36, slot 2026-10-19 09:00:00
14	2026-10-19 19:40:28	170	CANCELLED	Scheduled	Cancelled	Doctor 9, patient 6, slot 2026-10-17 11:30:00

Fig 12. IT6 · Total Cholesterol = 250 flagged HIGH

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	5	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Diagnostic Lab

Order #229 (Total Cholesterol) completed. Result is HIGH.

Pending tests	Waiting for sample	Waiting for result
5	4	1

Pending worklist

lab_order_id	appt_id	patient_name	ordered_by	test_id	test_name	category	unit	status	ordered_at	sample_col
12	16	Ravi Kumar	Dr. Rahul Verma	5	Fasting Blood Glucose	Biochemistry	mg/dL	Ordered	2026-10-19 16:40:28	None
13	16	Ravi Kumar	Dr. Rahul Verma	7	Total Cholesterol	Lipid Profile	mg/dL	Sample Collected	2026-10-19 16:40:28	2026-10-19
14	16	Ravi Kumar	Dr. Rahul Verma	9	HDL Cholesterol	Lipid Profile	mg/dL	Ordered	2026-10-19 16:40:28	None
15	12	Vijay Anand	Dr. Rahul Verma	21	ECG	Cardiac	None	Ordered	2026-10-19 18:40:28	None
230	171	Test Journey	Dr. Meera Iyer	1	Hemoglobin	Hematology	g/dL	Ordered	2026-10-19 19:41:18	None

1. Collect sample

Order

Order #12 | Ravi Kumar | Fasting Blood Glucose

2. Enter result

Order

Order #13 | Ravi Kumar | Total Cholesterol

Reference range: 0.00 to 200.00 mg/dL

Fig 13. IT7 · No invoice yet: 'Mark as paid' is not offered

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	6	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

Billing Desk | Analytics

Visit to bill

#171 Test Journey with Dr. Meera Iyer (19 Oct 09:00) [Not billed, 2 lab pending]

Only visits ready to bill

Test Journey

Cardiology · Dr. Meera Iyer · visit #171 on 19 Oct 2026 at 09:00

Item	Status	Amount (₹)
Consultation - Dr. Meera Iyer	-	850.00
Lab - Total Cholesterol	Ordered	300.00
Lab - Hemoglobin	Ordered	150.00

Expected total: ₹1,300.00

2 lab test(s) are not completed yet. The procedure will refuse to invoice this visit until the lab finishes. You can press Generate to show this.

Invoice

Generate invoice

No invoice yet. Generating one sums the consultation fee and all lab tests in a single transaction.

Recently paid

bill	Paid at	patient	doctor	Consultation ₹	Labs ₹	Total ₹	payment_mode
1	18 Oct, 18:50	Ravi Kumar	Dr. Arvind Menon	400.00	300.00	700.00	UPI
2	18 Oct, 17:50	Divya Sharma	Dr. Arvind Menon	400.00	0.00	400.00	Card

Fig 14. IT8 · Paid invoice still downloadable from Recently paid (Week 3 fix)

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
2	4	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Expected total: ₹1,630.00

3 lab test(s) are not completed yet. The procedure will refuse to invoice this visit until the lab finishes. You can press Generate to show this.

Recently paid

bill	Paid at	patient	doctor	Consultation ₹	Labs ₹	Total ₹	payment_mode
157	19 Oct, 19:42	Test Journey	Dr. Meera Iyer	850.00	800.00	1650.00	UPI
1	18 Oct, 18:50	Ravi Kumar	Dr. Arvind Menon	400.00	300.00	700.00	UPI
2	18 Oct, 17:50	Divya Sharma	Dr. Arvind Menon	400.00	0.00	400.00	Card
3	18 Oct, 16:50	Suresh Babu	Dr. Rahul Verma	900.00	1000.00	1900.00	Cash
4	18 Oct, 15:50	Lakshmi Narayanan	Dr. Rahul Verma	900.00	530.00	1430.00	UPI
5	18 Oct, 14:50	Aarav Patel	Dr. Karthik Subramanian	500.00	150.00	650.00	Card
6	18 Oct, 13:50	Mohammed Iqbal	Dr. Vikram Singh	700.00	900.00	1600.00	Cash
7	18 Oct, 12:50	Sneha Reddy	Dr. Farhan Sheikh	550.00	0.00	550.00	UPI
8	18 Oct, 10:50	Kavitha Raj	Dr. Sunita Rao	500.00	400.00	900.00	Cash
156	17 Oct, 16:40	Shalini Nair	Dr. Anita Desai	600.00	900.00	1500.00	UPI

Paid invoice

Bill #157 · Test Journey · ₹1,650.00

Download paid invoice

Fig 15. IT9 · Analytics, last 90 days: KPIs and charts from 05b history data

Deploy

Connected to **clinic_portal** on localhost

Today, 19 Oct 2026

Visits	Lab queue	Unpaid
1	1	1

Demo flow: Reception → Doctor Desk → Diagnostic Lab → Billing.

All SQL from this app uses parameterized queries (%s placeholders).

Digital Health Clinic & Diagnostic Lab Portal

Reception | Doctor Desk | Diagnostic Lab | Billing & Analytics

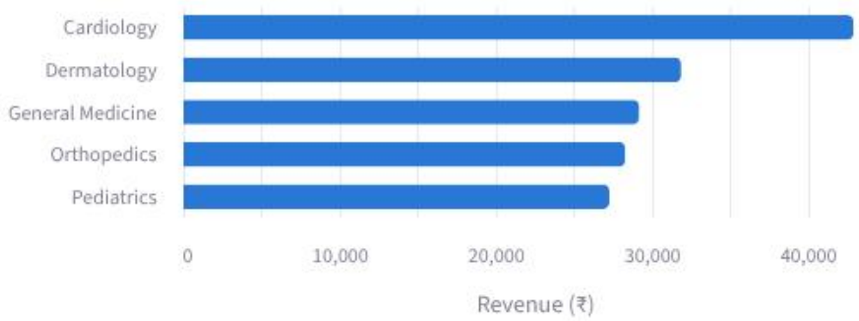
Billing Desk | Analytics

Period

Last 7 days | Last 30 days | **Last 90 days** | Last 365 days

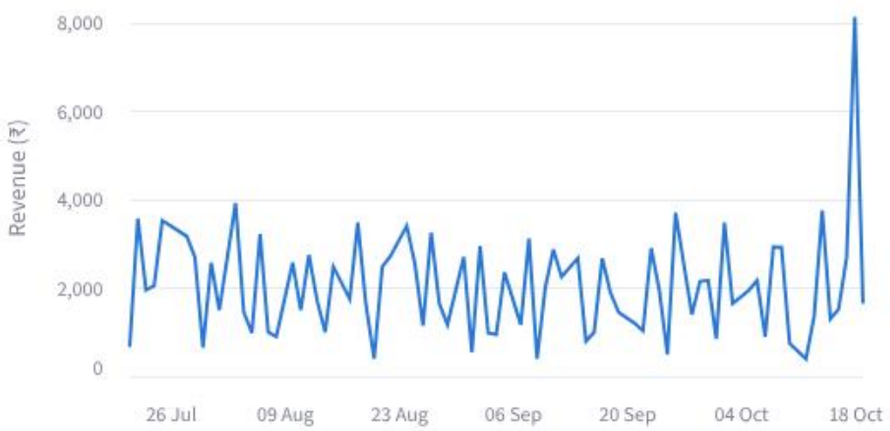
Revenue collected	Paid bills	Average bill	Visits	Outstanding (unpaid)
₹159,640.00	149	₹1,071.41	150	₹1,200.00

Revenue by department



› Show SQL

Revenue per day



› Show SQL

Most ordered lab tests



Busy hours

